

myCompiler

ex num-6

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FILES

main.m

Blocks.ai

Call Your self-hosted agent from anywhere. No inbound ports, no tunnel. Blocks.ai →

blocks.ai

Ads by EthicalAds

```

1  clc;
2  clear;
3  close all;
4
5  %% Parameters
6  lambda = 1;           % Wavelength
7  d = lambda/2;         % Antenna spacing
8  theta = -90:0.1:90;   % Angle scan range (degrees)
9  theta_rad = deg2rad(theta);
10
11 % Array sizes to analyze
12 N_array = [8 16 32 64];
13
14 beamwidth = zeros(1,length(N_array));
15
16 for idx = 1:length(N_array)
17
18     N = N_array(idx);
19
20     % Steering direction (broadside)
21     theta0 = 0;
22     theta0_rad = deg2rad(theta0);
23
24     % Array Factor Calculation
25     AF = zeros(size(theta_rad));
26
27     for n = 0:N-1
28         AF = AF + exp(1j*2*pi*(d/lambda)*n* ...
29             (sin(theta_rad)-sin(theta0_rad)));
30     end
31
32     % Normalize
33     AF_norm = abs(AF)/max(abs(AF));
34
35     % Compute 3-dB Beamwidth
36     mainlobe_max = max(AF_norm);
37

```

Run Save Output Input

=== Beamwidth Results (3-dB) ===

N = 8 : Beamwidth = 12.80 degrees

N = 16 : Beamwidth = 6.20 degrees

N = 32 : Beamwidth = 3.00 degrees

N = 64 : Beamwidth = 1.40 degrees

[Execution complete with exit code 0]

Beamwidth vs Array Size for ULA

Number of Antennas (N)	3-dB Beamwidth (degrees)
8	12.80
16	6.20
32	3.00
64	1.40